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Supertranslations as Soft Dressings

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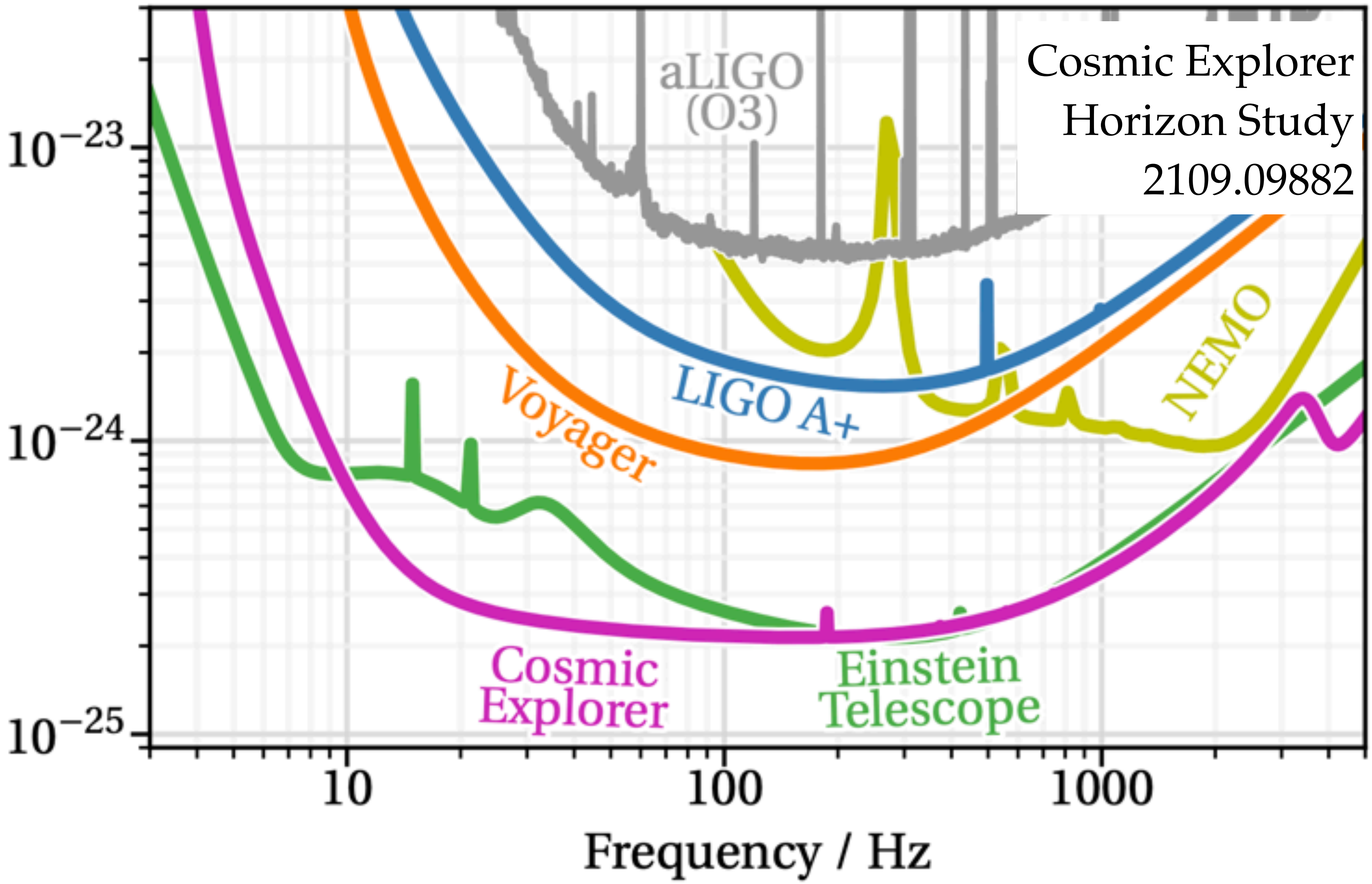
Motivation

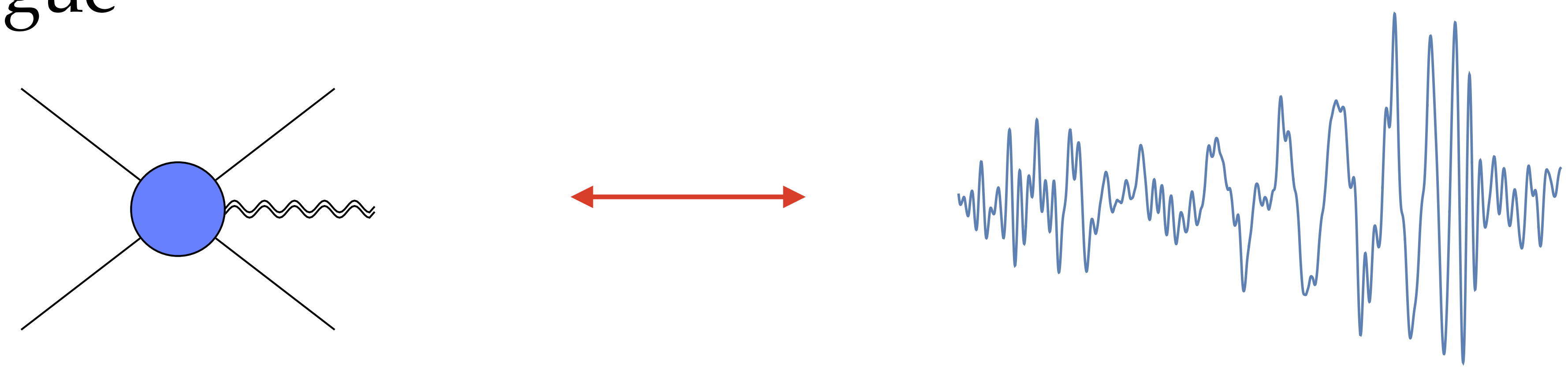
Quest for precision



Motivation

Learn about field theory → Thibault's talk

Fruitful dialogue



Waveform $\neq$ cross-section

IR / long distance properties?

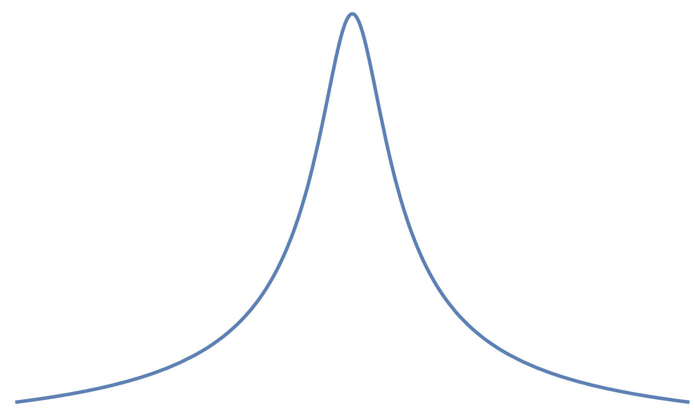
Outline

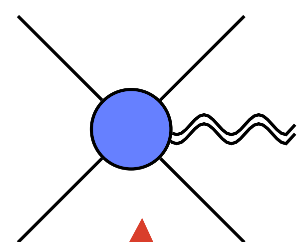
1. Waveforms
2. Large gauge transformations and “frames”
3. Single particle states
4. Supertranslations
5. Discussion

Waveforms

Waveforms

$\longrightarrow$ Fei's talk

$\mathrm{waveform}(t) =$ 

$= \dfrac{1}{\mathrm{distance}} \displaystyle\int$  $+ \cdots$

5 point amplitude

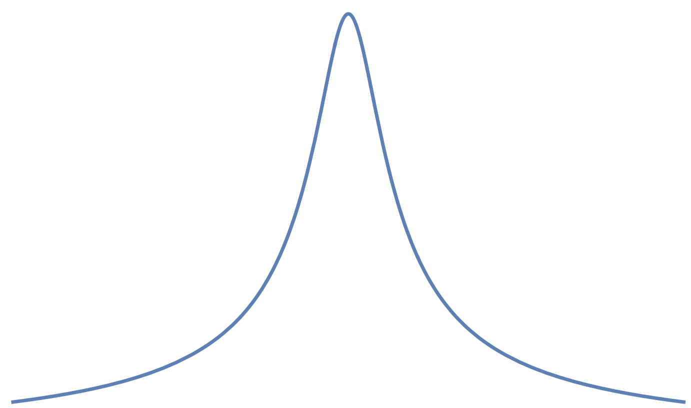
Classical point particle approximation (corrections at order G^7)

Kosower, Maybee & DOC "KMOC"

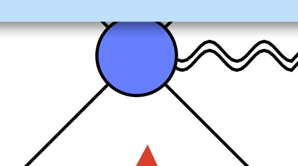
***Cristofoli**, Gonzo, Kosower & DOC*

Waveforms

→ Fei's talk

$\text{waveform}(t) =$ 

Useful: transfer precision information to bound case

$= \frac{}{\text{distance}} \int$  $+ \cdots$

5 point amplitude

Classical point particle approximation (corrections at order G^7)

Kosower, Maybee & DOC "KMOC"

Cristofoli, *Gonzo, Kosower & DOC*

Waveforms

Waveform: measuring metric change $g_{\mu\nu} - \eta_{\mu\nu} = h_{\mu\nu}$

QFT: observable is *expectation* of operator

$$h_{\mu\nu}(x) = \langle\psi|\mathbb{h}_{\mu\nu}(x)|\psi\rangle$$

Graviton quantum field

Waveforms

Waveform: measuring metric change $g_{\mu\nu} - \eta_{\mu\nu} = h_{\mu\nu}$

QFT: observable is *expectation* of operator

$$h_{\mu\nu}(x) = \langle\psi|\mathbb{h}_{\mu\nu}(x)|\psi\rangle$$

$$\mathbb{h}_{\mu\nu}(x) = \int \boxed{\mathrm{d}\Phi(k)}\, \varepsilon^{\eta}_{\mu}(k)\varepsilon^{\eta}_{\nu}(k)\, a_{\eta\eta}(k) e^{-ik\cdot x} + \text{h.c.}$$

Integral over physical momenta
$k^2 = 0$ "on-shell"

Waveforms

Waveform: measuring metric change $g_{\mu\nu} - \eta_{\mu\nu} = h_{\mu\nu}$

QFT: observable is *expectation* of operator

$$h_{\mu\nu}(x) = \langle \psi | \mathbb{h}_{\mu\nu}(x) | \psi \rangle$$

Graviton polarization tensor: helicity η

$$\mathbb{h}_{\mu\nu}(x) = \int \mathrm{d}\Phi(k) \boxed{\varepsilon^{\eta}_{\mu}(k)\varepsilon^{\eta}_{\nu}(k)}\, a_{\eta\eta}(k) e^{-ik\cdot x} + \mathrm{h.c.}$$

Waveforms

Waveform: measuring metric change $g_{\mu\nu} - \eta_{\mu\nu} = h_{\mu\nu}$

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$$h_{\mu\nu}(x) = \langle \psi | \mathbb{h}_{\mu\nu}(x) | \psi \rangle$$

$$\mathbb{h}_{\mu\nu}(x) = \int \mathrm{d}\Phi(k)\, \varepsilon^{\eta}_{\mu}(k) \varepsilon^{\eta}_{\nu}(k) \boxed{a_{\eta\eta}(k)} e^{-ik\cdot x} + \text{h.c.}$$

Graviton ladder
operator
helicity η

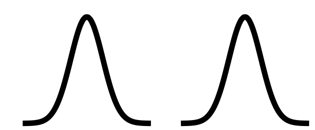
Waveforms

Initial state in classical approximation:

☑ Approximately localised

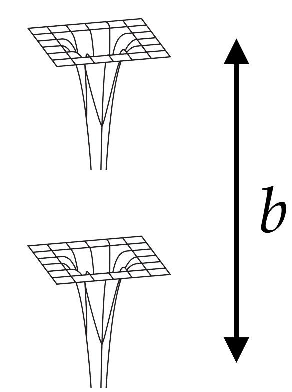
☑ Well-defined momentum

$$|\psi\rangle = \int \mathrm{d}\Phi(p_1, p_2)\, \phi(p_1)\phi(p_2)\, e^{ip_1\cdot b}\, |p_1, p_2\rangle \sim \int$$



$$e^{ip_1\cdot b}|p_1\, p_2\rangle$$

$$\sim$$



Waveforms

Initial state in classical approximation:

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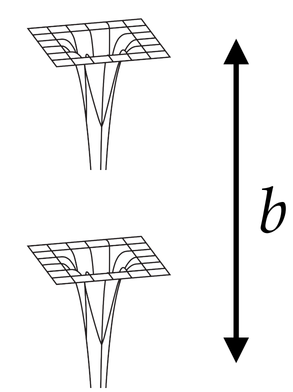
$\sim$

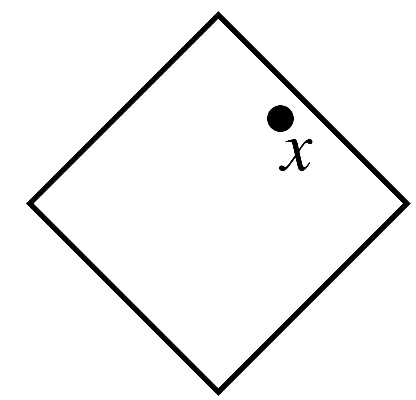


Note: no gravitons in initial state $a_{\eta\eta}(k)\,|\psi\rangle = 0$

Waveforms

Waveform: *future* expectation of metric operator

Time evolution operator: S matrix



$$\text{waveform} \equiv \langle\psi|S^\dagger \mathbb{h}_{\mu\nu}(x) S|\psi\rangle$$

Final state

Amplitudes!

Dominant term at large distances

(integral of Ψ_4)

$$|\psi\rangle \sim \int e^{ip_1\cdot b}|p_1\, p_2\rangle$$

Sets classical initial conditions

Kosower, Maybee & DOC "KMOC"

***Cristofoli**, Gonzo, Kosower & DOC*

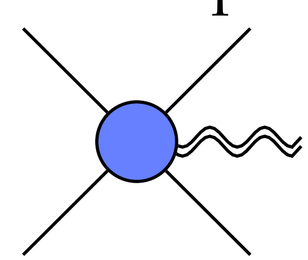
Waveforms

Waveform: *future* expectation of metric operator

$$\text{waveform} = \int \mathrm{d}\Phi(k)\,\left[\varepsilon^{\eta}\varepsilon^{\eta}\,e^{-ik\cdot x}\langle\psi|S^{\dagger}a_{\eta\eta}(k)S|\psi\rangle + \text{c.c.}\right]$$

$$= i\int \mathrm{d}\Phi(k)\,\varepsilon^{\eta}\varepsilon^{\eta}\,e^{-ik\cdot x}\boxed{\langle\psi|a_{\eta\eta}(k)T|\psi\rangle} + \text{c.c.} \qquad \text{@ LO}$$

5 point amplitude



$\longrightarrow$ Fei's talk

Waveforms

One oddity: *past* expectation of metric vanishes

$$\langle\psi|\mathbb{h}_{\mu\nu}(x)|\psi\rangle = \int \mathrm{d}\Phi(k)\, e^{-ik\cdot x}\varepsilon^{\mu}_{\eta}\varepsilon^{\nu}_{\eta}\;\langle\psi|a_{\eta\eta}(k)|\psi\rangle + \text{c.c.} = 0$$

No gravitons in initial state

Should we be worried?

Large Gauge Transformations & "Frames"

Frames

QFT creation operators

$$|k,++\rangle = a^{\dagger}_{++}(k)\,|0\rangle$$
$$|k,--\rangle = a^{\dagger}_{--}(k)\,|0\rangle$$

Two states
Propagating fluctuations

Formalism designed to compute radiative “angular” components

Frames

"Angular" components in gravity: Bondi shear

Angles θ^A

Shear

$$\mathrm{d}s^2 = g_{uu}\mathrm{d}u^2 + 2g_{ur}\mathrm{d}u\,\mathrm{d}r + 2g_{uA}\mathrm{d}u\,\mathrm{d}\theta^A + r^2\left(\Omega_{AB} + \frac{c_{AB}}{r}\right)\mathrm{d}\theta^A\mathrm{d}\theta^B$$

$$g_{rr} = g_{rA} = 0$$

Frames

"Angular" components in gravity: Bondi shear

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$$g_{rr} = g_{rA} = 0$$

Always available

Does metric of boosted Schwarzschild have zero shear?

Frames

"Angular" components in gravity: Bondi shear

Angles θ^A

Shear

$$ds^2 = g_{uu}\mathrm{d}u^2 + 2g_{ur}\mathrm{d}u\,\mathrm{d}r + 2g_{uA}\mathrm{d}u\,\mathrm{d}\theta^A + r^2\left(\Omega_{AB} + \frac{c_{AB}}{r}\right)\mathrm{d}\theta^A\mathrm{d}\theta^B$$

$$g_{rr} = g_{rA} = 0$$

Always available

Which Schwarzschild?

Does metric of boosted Schwarzschild have zero shear?

Frames

Boosted Schwarzschild

Kerr Schild "frame"

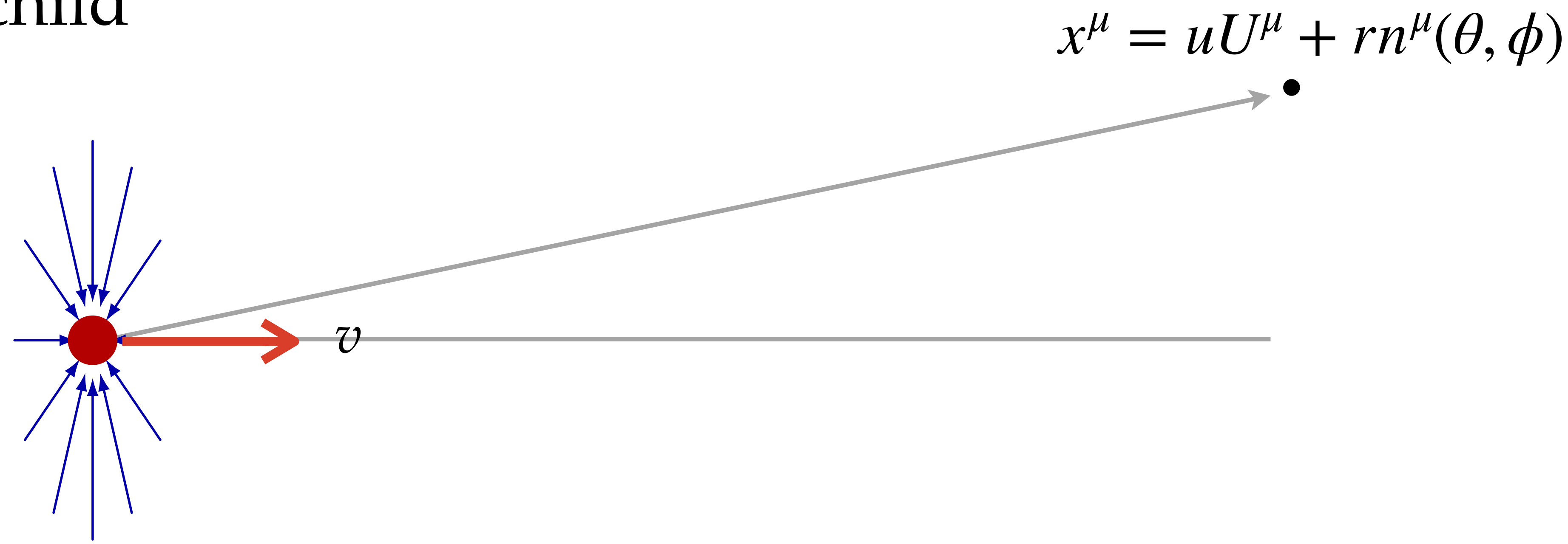
$$T = v \cdot x$$

$$R^2 = (v \cdot x)^2 - x^2$$

$$\mathrm{d}s^2 = \mathrm{d}s_0^2 - \frac{2GM}{R}\mathrm{d}(T - R)^2$$

Frames

Boosted Schwarzschild



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Frames

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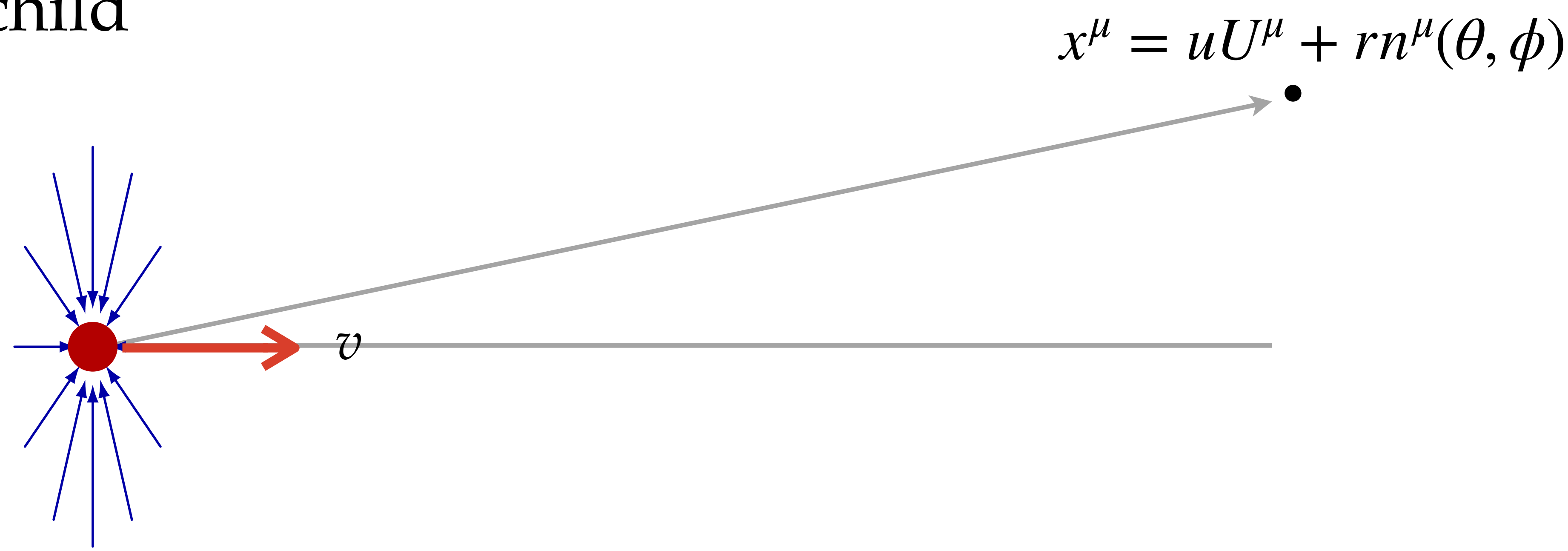


Kerr Schild "frame"

$$T = v \cdot x \;\rightarrow r(v \cdot n)$$

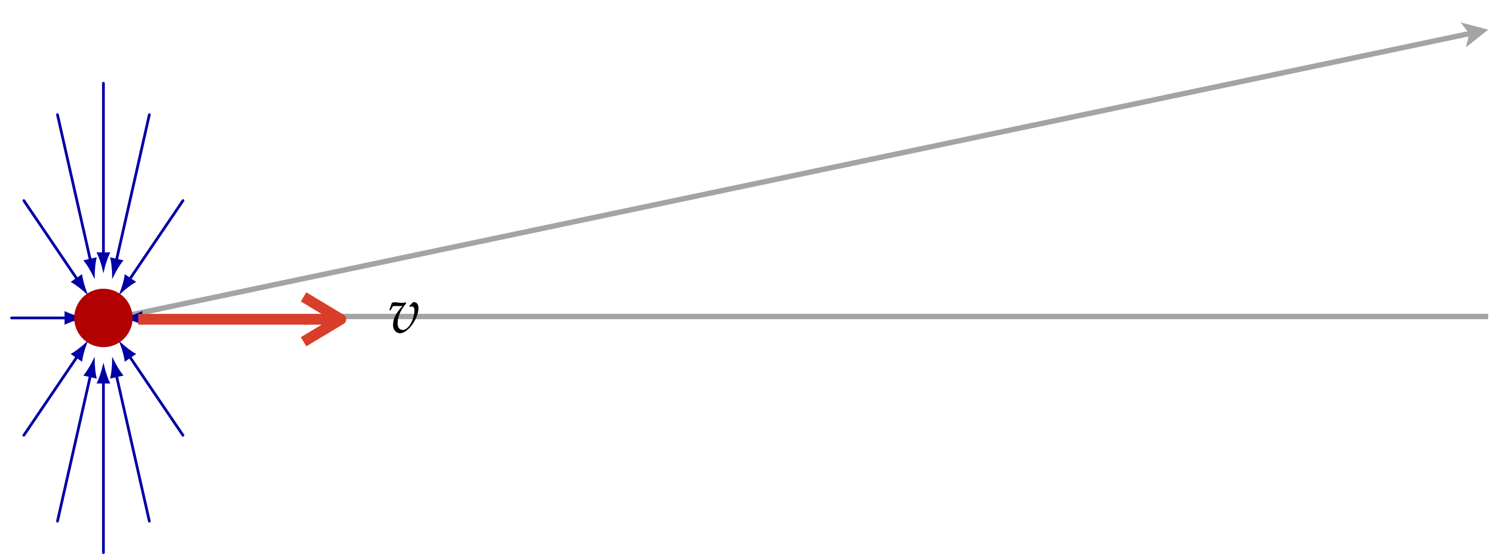
$$R^2 = (v \cdot x)^2 - x^2 \rightarrow r^2 (v \cdot n)^2$$

$$\mathrm{d}s^2 = \mathrm{d}s_0^2 - \frac{2GM}{R}\mathrm{d}(T - R)^2$$

Frames

Boosted Schwarzschild

$x^\mu = uU^\mu + rn^\mu(\theta, \phi)$



Kerr Schild "frame"

$$T = v \cdot x \ \to r(v \cdot n)$$

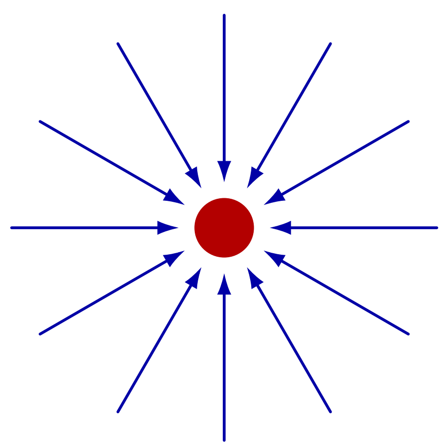
$$R^2 = (v \cdot x)^2 - x^2 \ \to r^2(v \cdot n)^2$$

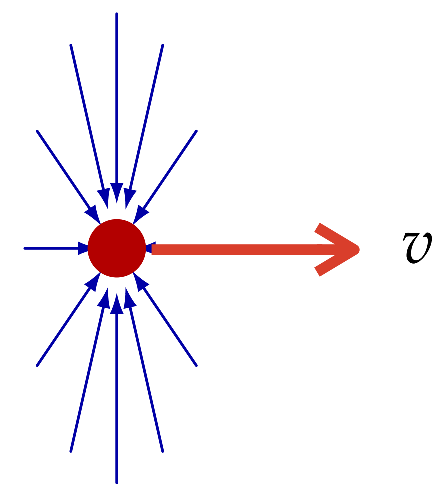
$$\mathrm{d}s^2 = \mathrm{d}s_0^2 - \frac{2GM}{R}\mathrm{d}(T - R)^2 \ \to \mathrm{d}s_0^2 - \frac{2GM}{r(v \cdot n)^3}\left(\mathrm{d}u^2 + \mathcal{O}(1/r^2)\right)$$

Poisson & Bonga

Frames

Boosted Schwarzschild





Kerr Schild "frame"

$$\boxed{\mathrm{d}s^2 = \mathrm{d}u^2 \left(1 - \frac{2GM}{r(v \cdot n)^3}\right) + 2\mathrm{d}u\,\mathrm{d}r + r^2 \left(\Omega_{AB} + \frac{0}{r}\right) \mathrm{d}\theta^A \mathrm{d}\theta^B + \cdots}$$

Automatically Bondi

Poisson & Bonga

Frames

Boosted Schwarzschild

Kerr Schild "frame"

No shear — persists for 2 BHs

$$\mathrm{d}s^2 = \mathrm{d}u^2\left(1-\frac{2GM}{r(v\cdot n)^3}\right) + 2\mathrm{d}u\,\mathrm{d}r + r^2\left(\Omega_{AB} + \frac{0}{r}\right)\mathrm{d}\theta^A\mathrm{d}\theta^B + \cdots$$

Poisson & Bonga

Frames

Boosted Schwarzschild

De Donder "frame"

$$\mathrm{d}s^2 = \mathrm{d}s_0^2 - \frac{4GM}{R}\left(v_\mu v_\nu - \frac{1}{2}\eta_{\mu\nu}\right)\mathrm{d}x^\mu\,\mathrm{d}x^\nu + \cdots$$

Frames

Boosted Schwarzschild

De Donder "frame"

$$\boxed{\mathrm{d}s^2 \to \mathrm{d}s_0^2 - \frac{4GM}{r\,n\cdot v}\left((n\cdot v\,\mathrm{d}r + r\,v\cdot\mathrm{d}n)^2 - \frac{1}{2}r^2\Omega_{AB}\mathrm{d}\theta^A\mathrm{d}\theta^B + \cdots\right)}$$

Not Bondi!

Frames

Boosted Schwarzschild

De Donder → Kerr Schild coords:

$$T \to T + 2GM \log R$$

Frames

Boosted Schwarzschild

De Donder → Kerr Schild coords:

$$\delta u = \sum_i 2GM_i\, n \cdot v_i \log n \cdot v_i + \text{const}$$

"Veneziano-Vilkovisky supertranslation"

Frames

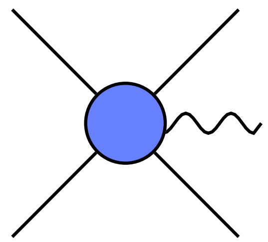
So naive amplitudes approach: Kerr-Schild gauge choice

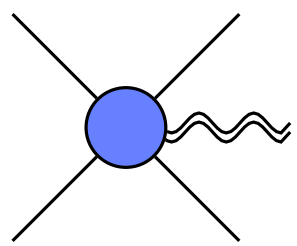
Kerr-Schild is canonical:

- Canonical BMS frame (zero shear)
- Canonical in QFT (simplest choice of ladder operators)

Frames

So naive amplitudes approach: Kerr-Schild gauge choice

$A_\mu(u) = \dfrac{1}{\text{distance}} \displaystyle\int$  $+ \cdots \quad \to F_{\mu\nu}(u), \quad$ invariant

$h_{\mu\nu}(u) = \dfrac{1}{\text{distance}} \displaystyle\int$  $+ \cdots \quad \to R_{\mu\nu\rho\sigma}(u), \quad$ *not* invariant

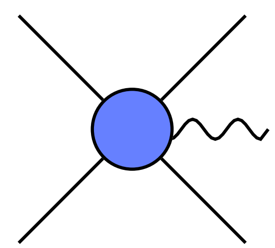
$$u \to u + 2Gm\, n\cdot v \log n\cdot v$$

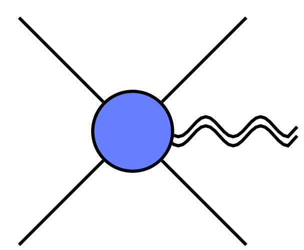
$\longrightarrow$ Thibault's talk

Bini, Damour, Geralico; Georgoudis, Heissenberg, Russo

Frames

So naive amplitudes approach: Kerr-Schild gauge choice

$$A_\mu(u) = \frac{1}{\text{distance}} \int$$  $$+ \cdots \quad \to F_{\mu\nu}(u), \quad \text{invariant}$$

$$h_{\mu\nu}(u) = \frac{1}{\text{distance}} \int$$  $$+ \cdots \quad \to R_{\mu\nu\rho\sigma}(u), \quad \textit{not} \text{ invariant}$$

$$u \to u + 2Gm\, n \cdot v \log n \cdot v$$

No local gauge invariant observables: $\longrightarrow$ Thibault's talk

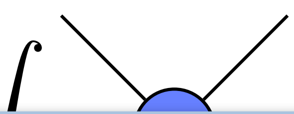
$$\psi_4(u) \to \psi_4(u + F(n))$$

Bini, Damour, Geralico; Georgoudis, Heissenberg, Russo

Frames

So naive amplitudes approach: Kerr-Schild gauge choice

$A_\mu(v$ $\quad 1 \quad \int$



1. How does this manifest at GW observatories?
2. What is the right "frame" = large gauge choice?
3. How can we adjust frame in QFT approach?

$h_{\mu\nu}($ … variant

$$u \to u + 2Gm\, n\cdot v \log n \cdot v$$

No local gauge invariant observables: → Thibault's talk

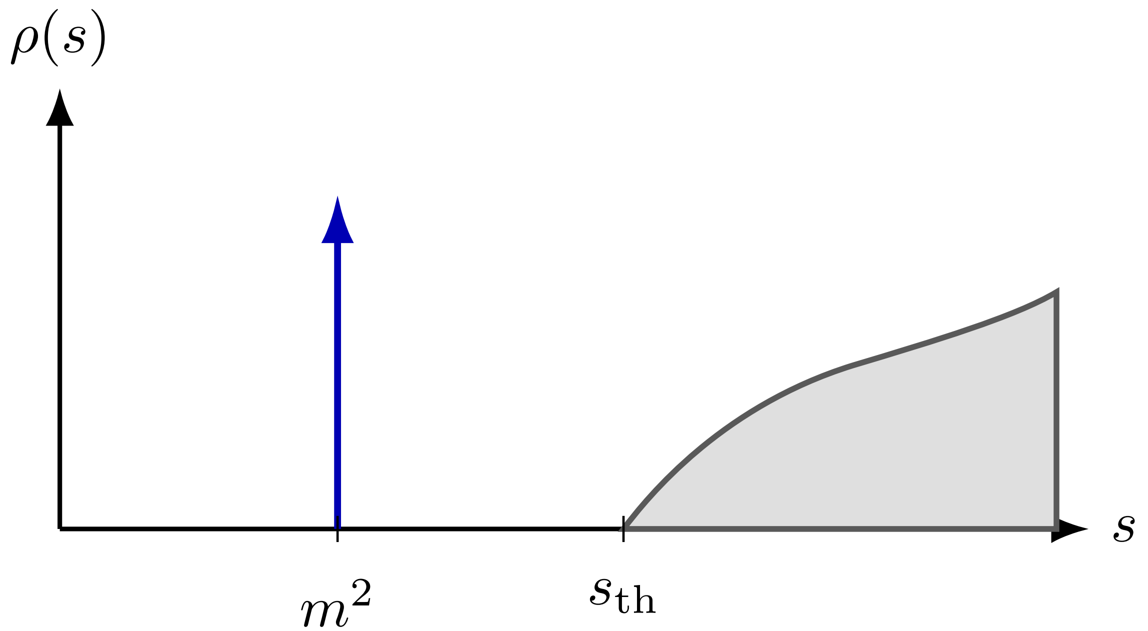
$$\psi_4(u) \to \psi_4(u + F(n))$$

Bini, Damour, Geralico; Georgoudis, Heissenberg, Russo

Single-Particle States

Single-Particle States

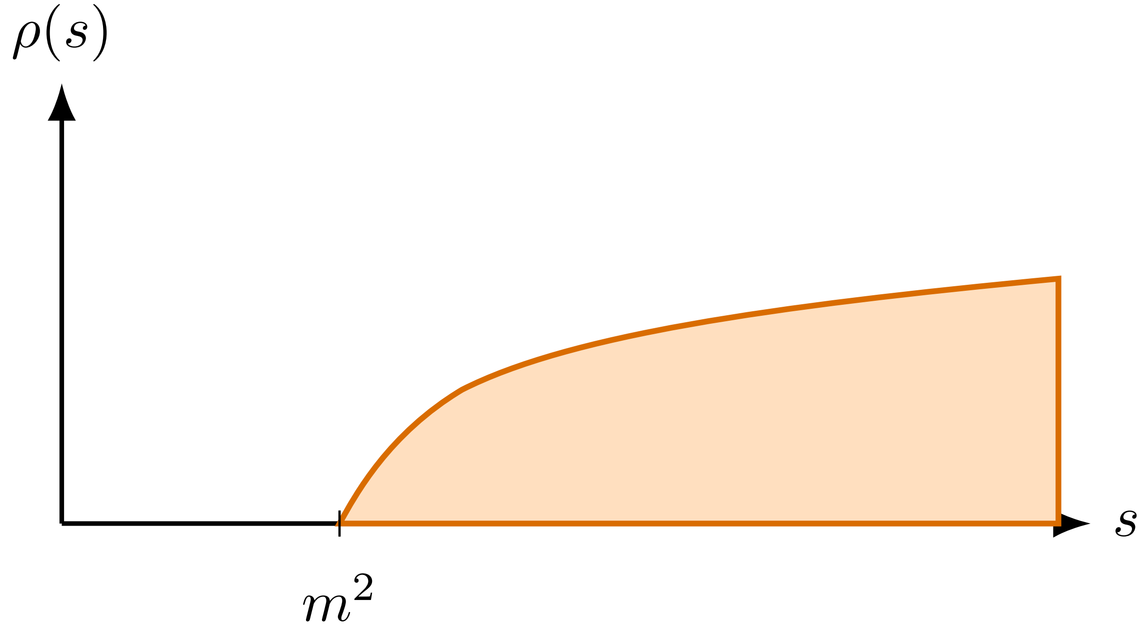
Gapped theory: isolated pole in spectral density



Massive pole well defined

Single-Particle States

No mass gap theory: no isolated pole



Massive pole *not* well defined

Single-Particle States

Definition of single particle state is ambiguous

Electromagnetic case
Simpler

$$|p\rangle \to \boxed{\mathcal{O}(a^\dagger_\eta)}|p\rangle$$

Creates photon "cloud"

Single-Particle States

Definition of single particle state is ambiguous

Electromagnetic case

Simpler

$$|p\rangle \to \mathcal{O}(a_\eta^\dagger)\,|p\rangle$$

Momentum of particle invariant

$$\mathbb{P}^\mu_{\text{tot}}\,|p\rangle = \boxed{\mathbb{P}^\mu_{\text{tot}}\mathcal{O}(a_\eta^\dagger)}\,|p\rangle$$

Cloud of *infrared* photons

Idealisation: photon momenta zero

Single-Particle States

Related idealisation

$$\text{waveform} = \int \mathrm{d}\Phi(k)\, \varepsilon^{\eta}\varepsilon^{\eta}\, e^{-ik\cdot x} \langle\psi|S^{\dagger}a_{\eta\eta}(k)S|\psi\rangle + \text{c.c.}$$

Radius *r* is very large: use stationary phase

$$\text{waveform} \simeq \frac{-i}{4\pi r} \int \mathrm{d}\omega\, \varepsilon^{\eta}\varepsilon^{\eta}\, e^{-i\omega u} \langle\psi|S^{\dagger}a_{\eta\eta}(k)S|\psi\rangle\Big|_{k=\omega n} + \text{c.c.}$$

$$x^{\mu} = rn^{\mu}$$

Single-Particle States

$$\text{waveform} \simeq \frac{-i}{4\pi r} \int \mathrm{d}\omega \, \varepsilon^{\eta} \varepsilon^{\eta} \, e^{-i\omega u} \langle \psi | S^{\dagger} a_{\eta\eta}(k) S | \psi \rangle \Big|_{k=\omega n} + \text{c.c.}$$

Justified provided $r\omega \gg 1$

Single-Particle States

$$\text{waveform} \simeq \frac{-i}{4\pi r} \int \mathrm{d}\omega \, \varepsilon^\eta \varepsilon^\eta \, e^{-i\omega u} \langle\psi| S^\dagger a_{\eta\eta}(k) S |\psi\rangle \Big|_{k=\omega n} + \text{c.c.}$$

Justified provided $r\omega \gg 1$

But…

$$\Delta h = h(\infty) - h(-\infty) = -i \int \mathrm{d}\omega \, \omega\delta(\omega) \, \tilde{h}(\omega) \neq 0$$

"Memory": waveform *must* include arbitrarily low frequencies

Idealisation: $1/r = 0$

Single-Particle States

Consider dressing with classical meaning

$$|p\rangle \to |p, f\rangle = e^{iQ_s[f]} |p\rangle$$

Infrared photons

$$Q_s[f] = \int \mathrm{d}\Phi(k) \boxed{\delta(\omega)} \big(f_\eta(n) a_\eta^\dagger(\omega, n) + \text{h.c.}\big)$$

$k^\mu = \omega n^\mu$

Defined by spin-weight ± 1 function

Larger class of localised, classical particle

$$|\psi\rangle = \int \mathrm{d}\Phi(p)\, \phi(p)\, |p, f\rangle$$

Single-Particle States

So what's changed?

1. Initial EM field has expectation value

$$\langle\psi|\mathbb{A}_\mu(x)|\psi\rangle = A_\mu(x) = \frac{1}{r}\sum_\eta \left(\varepsilon^\eta_\mu f_\eta(n) + \varepsilon^{\eta *}_\mu f^*_\eta(n)\right)$$

Spin-weight ± 1

Single-Particle States

So what's changed?

1. Initial EM field has expectation value

$$\langle\psi|\mathbb{A}_\mu(x)|\psi\rangle = A_\mu(x) = \frac{1}{r}\sum_\eta \left(\varepsilon^\eta_\mu f_\eta(n) + \varepsilon^{\eta *}_\mu f^*_\eta(n)\right)$$

Spin-weight ± 1

Observer velocity U^μ, gauge choice $\varepsilon^\eta(n)\cdot U = 0 \Rightarrow \eth = \varepsilon^+(n)\cdot\partial$

$$f_\eta(n) = \varepsilon^{\eta *}\cdot\partial F(n)$$

Single-Particle States

So what's changed?

1. Initial EM field has expectation value

Assume $F(n)$ is real

$$\langle\psi|\mathbb{A}_\mu(x)|\psi\rangle = A_\mu(x) = \frac{1}{r}\sum_\eta \left(\varepsilon^\eta_\mu \varepsilon^{\eta *}\cdot\partial F(n) + \varepsilon^{\eta *}_\mu \varepsilon^\eta \cdot \partial F^*(n)\right)$$

Observer coordinates $x^\mu = uU^\mu + rn^\mu$

$$A\cdot \mathrm{d}x = \mathrm{d}n^\mu\,\partial_\mu F(n) = \boxed{\mathrm{d}F(n)}$$

Pure gauge

Large gauge transformation

Single-Particle States

So what's changed?

2. Spectral density in Källén-Lehmann representation changes

$$i\Delta_F(p) = \int \mathrm{d}s \frac{\rho(s)}{p^2 - s + i\epsilon}$$

$$\rho(s) \to \rho_f(s)$$

Single-Particle States

Gravitational case

$$Q_s[f] = \int \mathrm{d}\Phi(k)\,\delta(\omega)\big(f_{\eta\eta}(n)a^\dagger_{\eta\eta}(\omega,n) + \text{h.c.}\big)$$

Spin-weight ± 2

$$f_{\eta\eta}(n) = \varepsilon^{*\mu}_\eta \varepsilon^{*\nu}_\eta \nabla_\mu \nabla_\nu F(n)$$

Expectation value, real F

$$\begin{aligned} h_{\mu\nu}\mathrm{d}x^\mu \mathrm{d}x^\nu &= \langle\psi|\mathbb{h}_{\mu\nu}(x)|\psi\rangle\,\mathrm{d}x^\mu\mathrm{d}x^\nu \\ &= r\,\mathrm{d}\theta^A\mathrm{d}\theta^B \boxed{(2\nabla_A\nabla_B - \Omega_{AB}\nabla^2)F(n)} \end{aligned}$$

BMS supertranslation $u \to u + F(n)$

Single-Particle States

Coherent state IR dressings:

- If real, generate pure gauge field expectations
- If imaginary: dual gauge transform (magnetic)

Classically associated with $u \to u + F(n)$

How does this emerge in the QFT approach?

Supertranslations

Supertranslations

$$\text{waveform} \simeq \frac{-i}{4\pi r} \int \mathrm{d}\omega\, \varepsilon^{\eta} \varepsilon^{\eta}\, e^{-i\omega u} \boxed{\langle\psi|S^{\dagger} a_{\eta\eta}(k) S|\psi\rangle}\Big|_{k=\omega n} + \text{c.c.}$$

All the time dependence

Expectation value to compute
Sensitive to dressing

Supertranslations

$$\text{waveform} \simeq \frac{-i}{4\pi r} \int \mathrm{d}\omega \, \varepsilon^{\eta} \varepsilon^{\eta} \, e^{-i\omega u} \boxed{\langle \psi | S^{\dagger} a_{\eta\eta}(k) S | \psi \rangle} \Big|_{k=\omega n} + \text{c.c.}$$

All the time dependence

Expectation value to compute
Sensitive to dressing

Focus on expectation:

$$\langle \psi | S^{\dagger} a_{\eta\eta}(k) S | \psi \rangle = \int \langle p_1' p_2' | e^{-iQ_s[f]} S^{\dagger} a_{\eta\eta}(k) S e^{iQ_s[f]} | p_1 p_2 \rangle$$

Short story: interactions same up to phase $e^{-i\omega F(n)}$

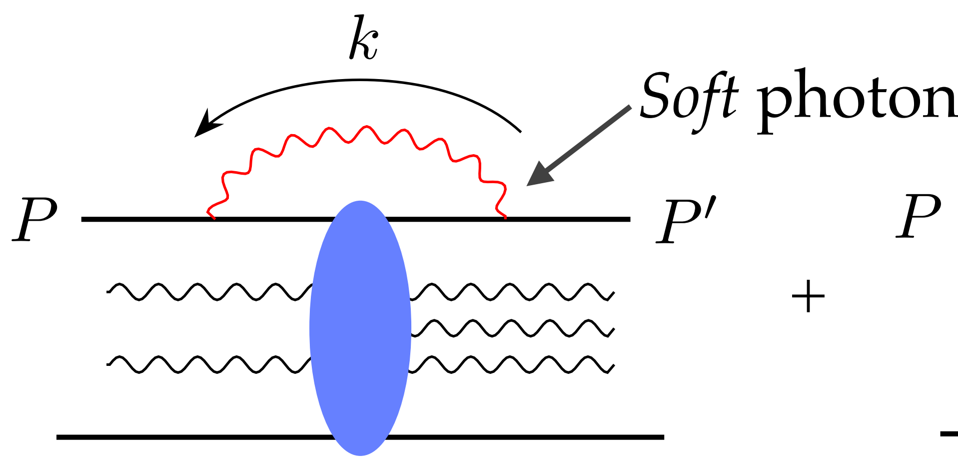
Supertranslations

Dressed expectation value

$$\langle p_1' p_2' | S^\dagger a(k) S | p_1 p_2 \rangle \to \langle p_1' p_2' | e^{-iQ_s} S^\dagger a(k) S e^{iQ_s} | p_1 p_2 \rangle$$

E&M:

$$[S, Q_s] =$$



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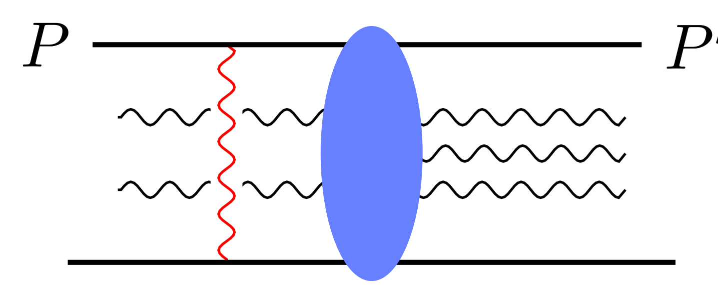


$+$

$$= -[S, Q_{h,M}] \qquad Q_{h,M} \sim \sum_i Q_{h,i} A_i^\dagger A_i$$

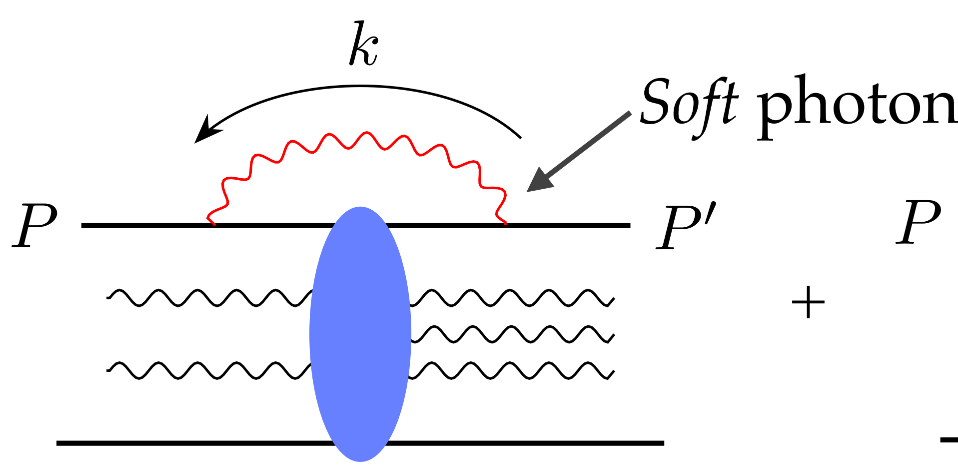
$$\langle p_1' p_2' | e^{-i(Q_s + Q_h)} S^\dagger a(k) S e^{i(Q_s + Q_h)} | p_1 p_2 \rangle$$

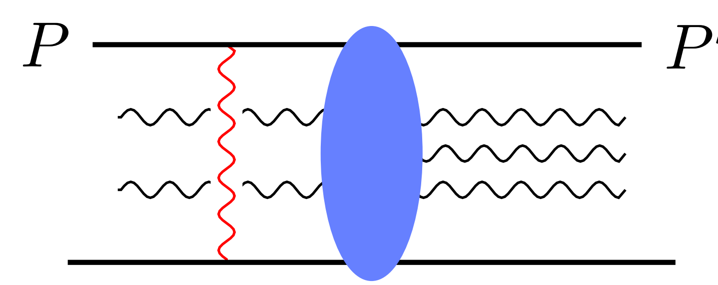
Supertranslations

Dressed expectation value

$$\langle p_1' p_2' | S^\dagger a(k) S | p_1 p_2 \rangle \to \langle p_1' p_2' | e^{-iQ_s} S^\dagger a(k) S e^{iQ_s} | p_1 p_2 \rangle$$

E&M:

$[S, Q_s] =$ 

$+$ 

$+$

$$= -[S, Q_{h,M}] \qquad Q_{h,M} \sim \sum_i Q_{h,i} A_i^\dagger A_i$$

$$\langle p_1' p_2' | S^\dagger e^{-i(Q_s + Q_h)} a(k) e^{i(Q_s + Q_h)} S | p_1 p_2 \rangle$$

Supertranslations

Dressed expectation value

$$\langle p_1' p_2' | S^\dagger a(k) S | p_1 p_2 \rangle \to \langle p_1' p_2' | e^{-iQ_s} S^\dagger a(k) S e^{iQ_s} | p_1 p_2 \rangle$$

E&M:

$$e^{-iQ_s} a(k) e^{iQ_s} = a(k) + \text{gauge}$$

So dressed EM waveform = "canonical" waveform + large gauge

$$A'^\mu = A^\mu_{\text{canonical}} + \partial^\mu F(n)$$

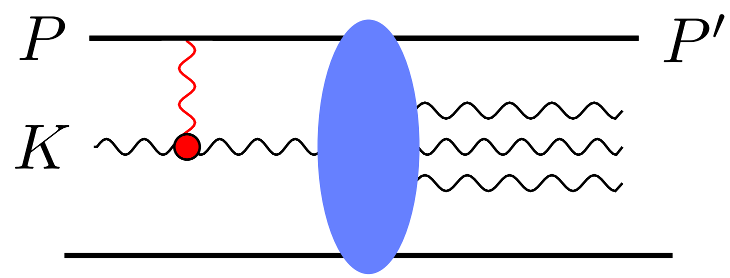
Supertranslations

Dressed expectation value

$$\langle p_1' p_2' | S^\dagger a(k) S | p_1 p_2 \rangle \to \langle p_1' p_2' | e^{-iQ_s} S^\dagger a(k) S e^{iQ_s} | p_1 p_2 \rangle$$

In GR, dressing involves $\eth^2 F(n)$

$$[S, Q_s] \supset$$



Loop integrals
remove ð!

$$Q_h = -\int F(n)\, a^\dagger(\omega n) a(\omega n) + Q_{h,i}$$

Supertranslations

Dressed expectation value

$$\langle p_1' p_2' | S^\dagger a(k) S | p_1 p_2 \rangle \to \langle p_1' p_2' | e^{-iQ_s} S^\dagger a(k) S e^{iQ_s} | p_1 p_2 \rangle$$

In GR, dressing involves $\eth^2 F(n)$

$$Q_h = -\int F(n)\, a^\dagger(\omega n) a(\omega n) + Q_{h,i}$$

$$\langle p_1' p_2' | e^{-i(Q_s + Q_h)} S^\dagger a(k) S e^{i(Q_s + Q_h)} | p_1 p_2 \rangle$$

Supertranslations

Dressed expectation value

$$\langle p_1' p_2' | S^\dagger a(k) S | p_1 p_2 \rangle \to \langle p_1' p_2' | e^{-iQ_s} S^\dagger a(k) S e^{iQ_s} | p_1 p_2 \rangle$$

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Supertranslations

Dressed expectation value

$$\langle p_1' p_2' | S^\dagger a(k) S | p_1 p_2 \rangle \to \langle p_1' p_2' | e^{-iQ_s} S^\dagger a(k) S e^{iQ_s} | p_1 p_2 \rangle$$

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$$= e^{-i\omega\, F(n)} a(k) + \text{Schw}$$

Supertranslations

Dressed expectation value

$$\langle p_1' p_2' | S^\dagger a(k) S | p_1 p_2 \rangle \to \langle p_1' p_2' | e^{-iQ_s} S^\dagger a(k) S e^{iQ_s} | p_1 p_2 \rangle$$

In GR, dressing involves $\eth^2 F(n)$

$$Q_h = -\int F(n)\, a^\dagger(\omega n) a(\omega n) + Q_{h,i}$$

$$\langle p_1' p_2' | S^\dagger \boxed{e^{-i(Q_s+Q_h)} a(k) e^{i(Q_s+Q_h)}} S | p_1 p_2 \rangle$$

$$= e^{-i\omega\, F(n)} a(k) + \text{Schw}$$

Combines with $e^{-i\omega u}$ to get $u \to u + F(n)$

Conclusion

One-one correspondence :

- BMS supertranslations of asymptotically flat spacetimes with N compact objects
- Real low-energy coherent dressings of states with N massive particles

Amplitudes setup: Kerr-Schild frame

Dressing also leads to shift energy-dependent shift of impact parameter